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Critical Approaches to Science & Philosophy

MARIO BUNGE
EDITOR

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*Deceased, 1961.

Introduction to the Transaction Edition

WHEN this book first appeared in 1964, Popper was little known outside a small circle of philosophers and scientists. The perception of his stature has greatly changed since then. He was knighted in 1965, and shortly thereafter a large international conference was held in London in his honor. That was when the memorable Popper-Carnap and Popper-Kuhn controversies took place. Since then Popper's views have been discussed by hundreds of scholars and quoted by thousands of scientists. For example, the quarterly *Philosophy of the Social Sciences* has been publishing regularly essays on Popper for over more than a quarter of a century.

Whether Popper's ideas have always been correctly understood is another matter. I doubt that they have, not only because he was always swimming against the current, but also because sometimes he changed his views without saying so. Moreover, he was seldom overly precise. Nor was he the prisoner of the *esprit de système*. Imprecision and unsystematicity always breed multiple interpretations, which in turn breed unending disputes. No wonder that his followers are divided on many important points of exegesis.

Undoubtedly, Popper has been the most influential philosopher of science and social philosopher of the second half of the twentieth century. His *Logic of Scientific Discovery* (1935, or 1934 if you belong to Popper's inner circle) and *The Open Society and Its Enemies* are modern classics. The former all but destroyed positivism, and the latter was the first scholarly study of the philosophical roots of totalitarianism left and right.

One of the reasons for Popper's popularity is that he wrote in a simple and beautiful style, always avoiding unnecessary jargon. (Joseph Conrad comes to mind.) Another reason is that he tackled inter-

esting problems, usually topical ones: he abhorred mental acrobatics as much as pomposity. A third reason is that, even when he was confused, Popper did no pussy-footing: he had strong opinions and expressed them bluntly—to the annoyance of experts in subtlety without substance.

For all these reasons, it is always a delight to read or re-read Popper, even when some of his views—or the lack of them in cases that call for definite views—may be irritating. For myself, I have been reading and re-reading him for over four decades.

I have also had the privilege of enjoying his friendship for a quarter of a century—until we split over the mind-body problem and the question of Popper's "world 3." I visited him and his wife Hennie several times, and we met at various scholarly meetings. Karl and Hennie Popper lived in the small town of Penn, south of London. They had a large and beautiful house, in the middle of a well-tended garden. The house was full of books, and Karl's manuscripts cluttered tables, chairs, and couches. (Karl had made much of the furniture himself, when his father insisted that everyone should have a manual trade.)

Hennie was not only Karl's life companion: she was also his muse and secretary. She was also a gracious hostess. She would not only feed us, but also serve me grape juice, knowing that it was my main addiction at the time. And, when the conversation slackened, she would prompt us, saying: "Go on. You may not have this opportunity again." Unlike Karl, who was a perfectionist, hence a bit of a procrastinator, she knew that life is short. Alas, hers turned out to be shorter than his.

Through the Poppers I got to know a number of members of their circle. Among them were Peter Medawar, Ernest Gombrich, Frederick von Hayek, Joseph Agassi, Imre Lakatos, and John Watkins. Karl and Hennie gave me and my wife Marta the first glimpse of the magnificent English theatre: they took us to Stratford upon Avon to watch *Hamlet*. But enough of reminiscences.

Because of the continuing interest of Popper's philosophy, it has been decided to reissue this volume, which was Popper's first Festschrift—and which pleased him greatly. This book helps one understand Popper's ideas, and it is full of interesting thoughts independent of them. It is also a window into the minds of the members of an extraordinary cast of characters, several of whom have earned the Nobel prize.

MARIO BUNGE
Foundations & Philosophy
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Preface

THIS collection of philosophical essays, written on four continents by scientists, philosophers, and humanists, is presented to Karl Raimund Popper on his sixtieth anniversary as a token of critical admiration and in recognition of his work, as well as in expectancy for his future contributions.

The author of *The Logic of Scientific Discovery*, *The Open Society and its Enemies*, *The Poverty of Historicism*, *Conjectures and Refutations*, and other important and controversial works, is acknowledged as one of the most original, penetrating, rigorous, and versatile thinkers of our time. He is also a very influential author, although most of the influence of his thought has been either indirect—namely, through the writings of his students and students of his work—or by way of reaction.

The depth of Popper's thought is related to its amazing range—an indication that nothing human is alien to him. Not unlike the philosophers of old, Popper has worked out and published his thoughts on the most varied subjects: logic, epistemology, metaphysics, physics, sociology, political science, and history. What is most remarkable is that he has not done it in a cursory and discursive way, but with the utmost rigor and an admirable clarity.

The astonishing clarity, depth, and range of Popper's thought, along with his caustic style, may not have helped him find followers prepared to agree with him on all counts. Perhaps many of those who tend to agree with Popper's philosophy of science will tend to disagree with Popper's social philosophy and vice versa. The important thing, however, is not that one would reject this or that idea of Popper's but that it is both instructive and stimulating to rethink what he has thought, whether agreeing with him or not. It is interesting and fruitful, though slightly dangerous, to argue with Popper, because he deals with deep problems in a way that is both careful and ingenious—a rare combination indeed. This is precisely one reason both Popper's followers and his intellectual opponents feel such a great respect for him: because he cannot help but stimulate

his readers and listeners. Sometimes this is true to the point of shock, particularly when he sets out to criticize prestigious doctrines or authors.

Criticism is one of Popper's favorite occupations. It is not just that Popper can *afford* to be a merciless critic because he is a creative thinker, but that his *attitude* to problems and their proposed solutions is critical. He knows that we never start from scratch but always from a certain cultural heritage which we must enrich by way of invention, discovery, and criticism. And he knows that invention and discovery, if they are to add to our rational understanding of the world and ourselves, must be criticizable, that is, testable—and criticized, that is, tested.

Of all intellectual endeavors, criticism is the most widespread and the least respected. Criticism can be applied to fact-gathering, to description, and to theory-building. It must be so applied if these activities are to be rational. Criticism, regarded as a minor or even shameful activity in some highly production-minded and success-oriented quarters, is the supervisor of creativity. Henri Poincaré believed that "criticism is the life of science," and Popper feels this is true of all intellectual work aiming at truth, as well as of all social endeavor aiming at freedom. Reason proper is constantly testing its own creations as well as those of intuition; and testing is criticizing.

The progress of knowledge—the perfecting of truths—is not a linear accumulation of definitive acquisitions but a zigzagging process in which counter-examples and unfavorable evidence ruin generalizations and prompt the invention of more comprehensive and sometimes deeper generalizations, to be criticized in their turn. A *critical approach* to problems, procedures, and results in every field of inquiry is, therefore, a necessary condition for the continuance of progress. One would even speak of a *critical method* for advancing knowledge, on any occasion but the present one. (Indeed, Popper's dislike of the word 'method', with the silly connotation of infallible recipe, is so intense that the writer heard him begin one of his lecture courses on scientific method by declaring that his subject did not exist.)

The title of this volume, then, is, in a sense, a homage to Popper's critical rationalism, critical empiricism, and critical realism. It is a tribute to his unceasing and uncompromising quest, not for final certainty, since he is a fallibilist, but for closer truth and increased clarity. Needless to say, ours is a critical homage.

No tribute to Popper could fail to be, at the same time, a homage to his wife Hennie, the closest of his friends and the most efficient of his collaborators. It is the writer's conviction that, were it not for Mrs. Popper, her husband would hardly cease answering letters and retouching—that is, criticizing—his manuscripts and galley proofs. In an important sense she is the coauthor of all of Popper's works and therefore deserves our gratitude.

The general recognition of the importance of Popper's work made it quite easy to compile this volume. In this task the editor had the valuable and friendly advice of Joseph Agassi, William W. Bartley III, Paul K. Feyerabend, Ernst Gombrich, Alan Musgrave, George Nadel, and John Watkins. Dr. Bartley is to be thanked for having compiled the biographical notes on the contributors to this volume, and Mr. Musgrave for having looked after Popper's bibliography. The editor's thanks are also due to the editorial staff of The Free Press of Glencoe.

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I

THE CRITICAL APPROACH
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CHAPTER 1

Rationality versus the Theory of Rationality

BY WILLIAM W. BARTLEY III

I

THE THREE PRINCIPAL PROBLEMS of philosophy are the problem of knowledge, the problem of rationality, and the problem of reconciling knowledge and rationality.¹

This essay will be concerned primarily with the second problem, to which it will present a solution. I will assume as correct most of the epoch-making contributions Karl Popper has made to the first problem, that of knowledge and its growth, which has been one of his principal philosophical interests. And I will use many of his remarks which bear more or less directly on the other problems. Thus the essay is in part an application of his philosophical thought; but it is also an attempt to interpret and generalize it, and to test and illustrate its power by extending it to further fundamental problems.²

¹ I shall use "intellectualism" to designate the movement of seventeenth-century rationalism, represented by philosophers such as Descartes and Spinoza, and reserve the words "reason," "rationality," and "rationalism" to refer broadly to that tradition whose members are dedicated to learning more through critical discussion. One of my main hopes will be to make this latter sense more precise. Needless to say, this view of rationalism is not at all dependent on the false notion that men can act rationally most of the time.

² Almost all other philosophical problems are directly related yet subordinate to the three fundamental ones. Of those not so related the most important are certain cosmological problems that may be deemed philosophical; and even these very often connect at least indirectly with the main three. For example, Descartes's most important cosmological theory—his identification of matter with extension—is directly sponsored by his views about the growth of knowledge.

The third problem, that of conflict between knowledge and rationality, typically arises when it is found that according to one's theory of rationality, knowledge is impossible; or that according to one's theory of knowledge, rationality is impossible. This conflict is usually occasioned by the existence of another conflict within the theory of rationality itself, a conflict that appears when it is discovered that according to one's theory of rationality, rationality is impossible. The problem of resolving the latter conflict I take to be the *problem of rationality* or of the *limits of rationality*.

My attempt to solve this problem will proceed in the following manner. First, in Section II, I shall indicate the kind of limitation of rationality I have in mind, specify the most important philosophical positions this limitation engenders, and summarize the claim and argument on which these positions rest. Then in Sections III and IV, I shall explain briefly why it is important to refute these positions by solving the problem, and shall outline my strategy for dealing with it. I shall proceed to consider, in Sections V–VII, the historical background behind the construction of theories of rationality, and will discuss critically two influential but unsuccessful specimen theories of rationality. In Sections VIII–X, as an historical hypothesis, I shall specify several important but unrecorded philosophical dogmas I believe are responsible for the perennial failure of theories of rationality. Finally in Sections XI and XII, I shall present an alternative theory of rationality which, freed of these dogmas, is able successfully to solve the problem of rationality and its limits.

II

There are, in the history of philosophy, a number of different senses in which rationality has been viewed as limited. On the one hand, some philosophers have maintained, as did Kant in some of his moods, that there are certain factual limitations connected with the nature of thought, or with the psychological and biological structure of the human mind. Or the limits of human reason may be thought to coincide with the limits of sense experience; or it may be argued that they coincide with the limits of science, which cannot provide answers to all questions that may be posed to it. Other philosophers have stressed certain physical conditions in nature, largely independent of man's psyche and its limitations, which make scientific investigation exceedingly difficult if not impossible in certain areas. The existence of radiation chaos,³ or the velocity of light, might be cited as limiting our research somewhat in certain parts of the universe.

Still other limitations are said to be connected with our historical existence and with the impossibility of predicting the historical future

³ See K. R. Popper, "The Aim of Science," *Ratio*, 1 (1957), p. 35.

because—among other reasons—we cannot predict the future growth of human knowledge.⁴ This limitation Popper himself has particularly stressed: we learn by refuting our present theories, by deriving predictions from them and trying to falsify those predictions; but we cannot derive or predict a refutation of these theories *from* these theories (provided they are consistent). Related limitations appear in physics and economics due to the existence of indeterminacy and “feedback.” Yet another limitation is connected with the necessarily selective character of description. Then there are the practical limitations of rationality such as those explored by Freud—due to human weakness, physical frailty, humanity.

In what follows, I shall not be at all concerned with any of these limitations. Rather, I shall deal with a so-called *logical* limitation of rationality, which is sufficient, without aid from other limitations,⁵ to perpetuate the aforementioned conflict between rationality and the theory of rationality.

The logical problem of rationality can be considered as that of defeating the two main philosophical positions that have grown from the claim that rationality is logically limited: *skepticism* and *fideism*. Paradoxically, although both positions are, I believe, inimical to philosophy, both are sheltered by similar philosophical arguments. To defeat these self-appointed guardians of the boundaries of reason it is necessary (1) to deal with a philosophical claim both make and (2) to refute the argument on which both rest their claim.

The *claim* is that *from a rational point of view, the choice between competing beliefs and positions and ways of life, whether scientific, mathematical, moral, religious, metaphysical, political, or other, is arbitrary.* In short, it is claimed to be demonstrable by rational argument that it is logically impossible to act and decide on rational grounds when it comes to such choices—even though the making of such choices in a nonarbitrary way can be considered to be the main task of rationality.

The core of the *argument* used by skeptics and fideists to back their claim consists in a simple analysis of what is commonly regarded as the rational way to defend ideas.⁶ The argument based on this analysis might

⁴ K. R. Popper, *The Poverty of Historicism*, Preface, pp. ix-xi.

⁵ This is not, of course, to deny that in individual philosophical discussions, the previously stated limitations—some of which have more to do with the limitation of knowledge than the limitation of rationality—have often been fused and even confused with the particular limitation that concerns me. Indeed, some writers who have advocated fideistic irrationalism, Pascal for example, have blended together many different kinds of rationality limitations in their writings—in order, for instance, to stress the weakness and misery of any man who insists on relying on his pitiful reason “without God.”

⁶ For a discussion of some related issues, see my article, “Achilles, the Tortoise, and Explanation in Science and History,” *British Journal for the Philosophy of Science*, May 1962.

be called the argument about the limits of rationality, the *tu quoque* or boomerang argument,⁷ the dilemma of ultimate commitment, the problem of ultimate presuppositions, or any number of other names. The argument is a commentary on the fact that any view may be challenged with such questions as "How do you know?" "Give me a reason," or even "Prove it!" When such challenges are accepted by the citation of further reasons which entail those under challenge, these may be questioned in turn. And so on forever. Yet if the burden of proof or rational justification can be perpetually shifted back to a higher-order premise or reason, the belief originally questioned is never effectively defended. In order to justify the original conclusion, it appears that one must eventually stop at something not open to question, for which one need not provide reasons when demanded. Usually, but not necessarily, the role of halting point of rational discussion is played by such things as standards, criteria, ends, or goals which one accepts irrationally in order to avoid infinite regress. But if to defend a position rationally is to give good reasons in justification of it, it would appear that this stopping point is not rationally defensible. If men conclude their rational justifying at different points, moreover, "ultimate relativism" arises; for some way of choosing rationally among competing ultimate stopping points by appeal to a common standard is excluded *in principle* by the way the problem is set. Even if all men did subjectively stop at the same place, the problem would remain of determining rationally whether this universal subjective stopping point led to objectively true statements about the world.

Skepticism and fideism, then, share both the claim and the powerful argument which underlies it. They differ in their opposed practical attitudes toward the claim. Whereas the skeptic suspends his judgment about competing positions, or so he says, the fideist makes an irrational commitment to one or another of them, or to some authority or tradition claiming to possess the competence or the right to make such decisions for him. The claim and the argument are, however, far more basic features of the two positions than are the different concluding attitudes they choose: a position that cannot escape skepticism cannot refute fideism.

III

Philosophical incapacity to answer the fideistic and skeptical claim and argument has some serious consequences. In the first place, the argument, if correct, implies that it is pointless from a rational point of view for men to argue rationally about their extremely different "ultimate presuppositions" or commitments. T. S. Eliot, who found a way to become a

⁷ The name "boomerang argument" was suggested to me by Professor Hans Albert.

fideist Christian when he had learnt why his teacher Bertrand Russell could not refute skepticism, embraced such an implication in *After Strange Gods*, in which he wrote :

I am not arguing or reasoning or engaging in controversy with those whose views are radically opposed to such as mine. In our time, controversy seems to me, on really fundamental matters, to be futile It requires common assumptions.

So it would seem that the limits of rational argument within any ultimate philosophical position are defined by reference to that object or belief in respect to which commitment is made or imposed, in regard to which argument is brought to a close. And if, since the limitation is a logical one, all men share it, if *no one* can escape irrational commitment, then no one can be criticized rationally for having made such a commitment, no matter how idiosyncratic. Bertrand Russell chose to commit himself to the so-called scientific principle of induction, and T. S. Eliot chose Anglo-Catholicism; and that is all there is to it, so it is often argued.

This means that any irrationalist has a *rational excuse for irrationalism* and a secure refuge from any criticism of this irrational commitment. The irrationalist can reply "*tu quoque*" to any critic, and remind him that people whose own rationality is similarly limited should not berate others for admitting to and acting on the limitation.

On the other hand, it should be noted that however useful his rational excuse for irrationalism may seem, the irrationalist pays a high price for using it. The strength and survival of the *tu quoque* argument does indeed make rational criticism of irrationalist commitment boomerang. But the *tu quoque* argument itself also has a boomerang effect on those who use it. I have discussed this point at some length elsewhere, with examples, and wish only to suggest the character of this backfire here by quoting a brief passage from my discussion :⁸

To the extent that anyone employing [the *tu quoque*] strengthens his own position by insuring that it is parallel to his opponent's, to that extent he increases the invulnerability of the *opponent* to criticism. For the opponent, if criticized, may also use the *tu quoque*. Those who gain a refuge of safety for themselves through appeal to the limits of rationality thereby provide a similar refuge for all others whose commitments differ from theirs. Thus,

⁸ See W. W. Bartley III, *The Retreat to Commitment* (New York: Alfred A. Knopf, Inc., 1962), Chap. IV, section 2, "How Shoes Can Be Danced Into Holes," pp. 95-104. In this book I have applied and developed some aspects of the present arguments in a less general way in terms of a specific case study of the development and stresses of American Protestant religious thought in the past century, particularly in its turn from "Protestant liberalism" or "religious modernism" to "neo-orthodoxy." See also my discussions in my dissertation for the Ph.D., "Limits of Rationality: A Study of Some Logical Problems in Contemporary Pragmatism and Related Movements," University of London Library.

the many criticisms which [irrationalists] have leveled at rationalism and liberalism become as pointless as those the liberals have directed at [irrationalism]. Ultimately, the use of the *tu quoque* makes nonsense of the idea of the historical development and change of ideas in the face of criticism. . . . In sum, the belief that rationality is ultimately limited, by providing an excuse for irrational commitment, enables . . . any . . . irrationalist to make an irrational commitment without losing intellectual integrity. But at the same time, anyone who makes use of this excuse may not, in integrity, criticize the holder of a different commitment. One gains the right to be irrational at the expense of losing the right to criticize. One gains immunity from criticism for one's own commitment by making any criticism of commitments impossible.⁹

IV

If the cost of his excuse makes it hard for the irrationalist to live comfortably with it, this does not reduce the argumentative effectiveness of the excuse on his opponent. Is the rational excuse for irrationalism really flawless from a rational point of view? Or can the excuse be defeated with rational arguments?

Before attempting to answer these questions, I would like to make the task more manageable; and without distorting any issues or minimizing any difficulties. At the same time I want to focus the skeptico-fideistic argument as sharply as possible in a single example by considering some specific tradition, philosophical viewpoint, or way of life of our culture to see whether irrational commitment is really required of its adherents. Some necessary care in the choice of a tradition to be examined will, I think, satisfy both these desires. The tradition I propose to study briefly is the rationalist tradition itself. Although this is very much a way of life in its own right, with its own history, its proponents rarely consider it historically and critically. Since, in any case, our main problem is about the limits of rationality in the choice of competing ways of life, we might turn the problem on itself by asking *what are the limits of rationality when it comes to choosing between the rationalist and some other way of life?* Must the rationalist, for instance, make an irrational commitment to reason? If so, would this not preclude him from rationalism, a tradition which has taught that everything one accepts must be rationally defensible? Is a real rationalist, then, even *possible* theoretically?

In dealing with the development of the rationalist tradition, its triumphs and vicissitudes, and with the growth and occasional decay of rationality within it, I have found helpful the two concepts of "crisis of

⁹ *The Retreat to Commitment*, pp. 96-97, 103.

identity" and "crisis of integrity."¹⁰ In their technical psychological sense, these terms are meant to refer to two turning points in the lives of most people. The problem of identity, usually encountered in late adolescence and early adulthood, is to mold out of one's heritage, one's conception of oneself, and one's conjectures about the image one presents to others, some satisfactory self-image, identity, or personal identification. The problem of integrity, which typically occurs in later life, although it is often fused with the problem of identity in the life of the *homo religiosus* or *philosophicus*, is this: Given one's identity, purpose, and claims about oneself, how can one live up to them, and cope with the fact that one can rarely if ever be entirely successful in such aims?

My thesis is that the perpetual crisis of integrity into which rationalists are continually falling or being forced is due to a neglected crisis of identity in the rationalist tradition; neglected partly because of a general failure on the part of philosophers to make deliberate efforts to develop a theory of rationality as well as a theory of knowledge. Because of these crises, the valuable handy man in the house of irrationalism—the *tu quoque*—is the skeleton in the cupboard of rationalism. Rationalists are overcommitted to a notion of rationality, or rationalist identity, that is impossible to attain; and the inevitable frustration of the effort to satisfy this overcommitment prevents them from achieving integrity. At the same time, the failure of the rationalist tradition to resolve its crisis of integrity enables many irrationalists, whatever their affiliations, to preserve their own identities without loss of integrity.

My attitude toward this situation is suggested by an anecdote about Gottlob Frege that Popper enjoys telling occasionally in his lectures. When Frege heard about Russell's discovery of paradoxes in his own and Frege's theories, the latter cried out: "*Die Arithmetik ist in's Schwanken geraten!*" (Roughly, "Arithmetic has been set spinning!") In fact, it was not arithmetic, but Frege's theory of arithmetic, that went spinning in vicious circles. Such mistakes Popper has dubbed (doubtless a bit unfairly to Frege) examples of "Frege's Mistake." A similar mistake is made about rationality and the possibility of being a rationalist. The blame for continued failure by rationalists to answer skeptical and fideistic arguments should, in fact, be placed on the inadequacy and primitive character of our *theories of rationality* or on our conception of rationalist identity, rather than on our rationality or reasoning capacity itself, where Pascal, Kant, and many others have put it. It is not the miserable state of the human creature that sponsors irrationalistic fideism and skepticism, but

¹⁰ I use these terms in a sense I borrow from the noted sociologist and psychoanalyst Erik H. Erikson, of Harvard University. See his *Childhood and Society* (London: Imago Publishing Company, 1950); *Young Man Luther* (New York: W. W. Norton & Company, 1958), p. 14 *passim*; and *Identity and the Life Cycle, Psychological Issues*, Monograph 1, 1959. See also my discussion of Erikson in *The Retreat to Commitment*, pp. 3, 10, 59, 218.

the miserable state of the philosophical theory of rationality that that human creature has accepted.

To illustrate this thesis I shall single out three different answers to the problem of rationality, three stages in the quests for identity and integrity in the rationalist tradition. Needless to say, these stages are not to be interpreted as exhaustive. They are *comprehensive rationalism*, *critical rationalism*, and *comprehensively critical rationalism*.

The history of modern philosophy is largely that of the failure of *comprehensive rationalism* to defeat skepticism and fideism. Most contemporary philosophies are forms of *critical rationalism*. *Comprehensively critical rationalism*, the third stage of this triad, is my attempt to weave together the comprehensive aims of comprehensive rationalism and the critical spirit of critical rationalism—and thereby to preserve the good intentions of the two earlier concepts while avoiding their fatal difficulties. Although my theory about rationalist identity rejects most traditional and contemporary characterizations, I believe it can both solve the main problem in response to which theories of rationalist identity arose and avoid those aspects that engendered the crisis of integrity in rationalism.

V

How did theories of rationalist identity originate? And which of their tenets or assumptions brought them to defeat before the argument about the limits of rationality?

The problem of rationality, and of rationalist identity, has been the standard and controlling predicament of philosophy since the late sixteenth century, although its origins are much earlier and its classical developments considerably later. It had already been discussed extensively in antiquity; and during the Hellenistic period it became for a short time dominant in philosophical discussion, its most detailed statement at that period having been made by Sextus Empiricus. For understandable reasons, the problem became centrally important again in modern intellectual history during the Renaissance and Reformation, when for the first time since the secular decline of Rome men were forced to choose among a great number of radically competing views. Identical arguments about the limits of rationality could be used skeptically by Protestants, Humanists, and scientists to attack the traditional authority and intellectual pretensions of the Roman Catholic Church; and fideistically by the Roman Catholics to attack the Protestant and scientific rejection of the authority of the Pope and of Aristotle in the name of reason; and again fideistically by the Protestants to answer Roman Catholic allegations that Protestantism led inevitably to skepticism. Similar arguments about the limits of rationality were widely quoted in support of fideism and skepticism during

Descartes's lifetime, and importantly influenced the emphases and themes of his philosophical program. Indeed, his thought can hardly be understood without some acquaintance with the currents of skeptical thought in European philosophy flowing not only from the religious and political upheavals and the scientific revolution, but also from such events as the Latin translation and publication in 1562 of the *Hypotyposes* of Sextus Empiricus, whose ideas were widely disseminated during the following century.¹¹

At their modern origins, theories of rationality served a strong utilitarian purpose. It was important for those who challenged traditional ecclesiastical and political authorities to be able to show that disputes could nevertheless still be settled in an orderly way; that traditional political, religious, and intellectual authorities, having come to seem rather arbitrary and irrational, could be displaced without producing social anarchy and intellectual chaos since they would be replaced by the authority of reason. The various schools of modern philosophy arose in an attempt to adjudicate among competing views by providing *rational* authorities to substitute for unwanted forms of traditional and hereditary authority. Since then, almost every "philosophical revolution," each being a phase in the search for rationalist identity, has disclosed that the previous candidate for intellectual authority was unsatisfactory and has proposed a new, supposedly more satisfactory, rational authority. The church was to be replaced by intellectual intuition, intellectual intuition by sense experience, sense experience by a certain language system, and so on. The story is almost always the same: past philosophical error is to be given a positive explanation by attributing it to the acceptance of the guidance of a false rational authority. Where these successive revolutions are fundamentally defective—apart from such assumptions as that error rather than knowledge needs to be explained—is in a depressingly similar structural pattern they share, which predestines them to failure.

VI

The traditionally dominant, and perhaps still most common, conception of rationalist identity—comprehensive rationalism—can be traced back at least to Epictetus, who wrote in his *Discourses* (Chapter II) that "To be a reasonable creature, that alone is insupportable which is unreasonable; but everything reasonable may be supported." Such conceptions combine two main requirements for rationalist identity.

¹¹ Sextus Empiricus, *Works* (Cambridge, Mass.: Harvard University Press), 4 vols. See also Richard H. Popkin: *The History of Scepticism from Erasmus to Descartes* (Assen: Van Gorcum; 1960).

1. In the first place, a rationalist accepts any position that can be justified or established or supported by appeal to the rational criteria or authorities.
2. In the second place, a rationalist accepts *only* those positions that can be justified in this way.

In their highly trite and stylized way, histories of modern philosophy usually say nothing about theories of rationality taken as a whole, but instead focus their attention on several basically subordinate questions which arise only when comprehensive rationalism is assumed as correct. Of these subordinate questions, the most important has probably been: "What is the nature of the rational authority or criterion to which a rationalist appeals to justify all his opinions?" The well-known modern theories of knowledge are usually functions of the answers philosophers have given to the question, and fall into two main categories:

1. According to the intellectualists (or Rationalists in the seventeenth century sense), the rational authority is to be found in the intellect or Reason. A rationalist justifies his beliefs by appealing to intellectual intuition.
2. According to empiricists, the rational authority lies in sense experience. An empiricist justifies his beliefs by appealing to sense observation.

There is no need to tell the familiar story of the failure of both these approaches. To use Popper's phrase, they were "too narrow and too wide."¹² Both intellectualism and empiricism endorsed as rational some obviously irrational theories (they were too wide), and excluded as irrational some theories that seemed highly rational (they were too narrow). Kant, with his antinomies, showed that intellectualism was too narrow; and Hume and others brought to light the narrowness of sense experience, which could not even provide a rational endorsement for so great an achievement as Newton's dynamics.

Although a recalcitrant comprehensive rationalist might diffidently admit his failure to meet such objections and thus far to have provided an adequate version of rationalism in which all rational beliefs could be justified and all irrational ones excluded, some such person might insist that this indicated no more than temporary bad luck, to be followed by the eventual triumph of some kind of comprehensive rationalism. That such optimism would be misguided can be seen in the fact that the two requirements for comprehensive rationalism cannot be held simultaneously. If we take the second requirement seriously, we will have to justify the first; but this cannot be done by sense experience, intellectual intuition, or any other rational authority ever proposed. Moreover, any such justification

¹² See Popper, "The Demarcation Between Science and Metaphysics," in *Conjectures and Refutations* (New York: Basic Books; 1963).

of the practice of accepting the results of argument, even if it could be achieved, would be pointless unless it were accepted in advance that a justification should be accepted at least here—which is just what is in part at issue. The argument would be acceptable and convincing only to those who had already adopted the principle that arguments should count.

As a consequence, the first and second requirements cannot be maintained simultaneously, for the second forbids the holding of any unjustifiable principles. This means that at least one of the requirements must be rejected. Surely this must be the second; for it can be shown by independent argument to be self-contradictory since it too cannot be justified by appeal to the rational criteria or authorities. Therefore the requirement asserts its own unacceptability: if it is true, it must, according to its own stipulations, be rejected.¹³

Such a collapse of comprehensive rationalism would seem to strengthen the position of the fideist, and has in fact done so historically. If the so-called rationalist not only cannot justify his own presuppositions rationally, if his position is basically self-contradictory, an irrationalist need hardly worry if his commitments are unjustifiable. He may really seem better off, since he did not claim to be able to justify them rationally, and he therefore surpasses the rationalist in integrity. Hence, when a rationalist accuses another of irrationalism, the irrationalist can reply that what is impossible cannot be morally demanded, and that the sort of comprehensive rationalism which the rationalist demands is, *rationally speaking*, impossible.

VII

That such difficulties have battered rationalists severely, from antiquity to the present, and have even driven some, if reluctantly, into outright irrationalism, can hardly be doubted. Many rationalists, however, have attempted to face and dispose of their difficulties without abandoning rationalism. Their responses to the problem, crude and sophisticated alike, amount to quests for some alternative theory of rationalist identity that might escape the quandaries of a primitive comprehensive rationalism.

These attempts vary widely in substance as well as sophistication. Here I propose to examine one prominent representative of what I regard as the most influential contemporary approach to the theory of rationality, which

¹³ Some of the components of my argument here are taken from Popper, *The Open Society and Its Enemies* (Princeton: Princeton University Press, 1950), Chap. XXIV, p. 416 *passim*. I believe that Popper's formulation, which shows that the position—by asserting its own falsity—is self-contradictory in a way analogous to the original statement of the liar paradox (which was, strictly speaking, nonparadoxical), is an important improvement on previous attempts to refute comprehensive rationalism.

I shall give the name of "critical rationalism."¹⁴ In applying this name to a number of philosophers who differ widely among themselves on important philosophical issues, I suggest only that their approaches to skepticism and fideism, and to the problem of the limits of rationality, share some common fundamental characteristics. The representative I have chosen to illustrate such an approach is A. J. Ayer, who, in 1956—having earlier abandoned the comprehensive rationalism of sense experience he championed in *Language, Truth and Logic* (1936)—presented a variety of critical rationalism in *The Problem of Knowledge*. Along with Morton White's *Toward Reunion in Philosophy*, published in the same year, Ayer's statement provides a most perceptive and self-conscious development of the general approach—and one which is lucid and precise enough to make the implications of the position clear. Ayer's account has the added merit of being explicitly devoted to considering and answering the skeptic's claims.

Ayer's strategy is to minimize the importance of the skeptical arguments while granting their cogency, to make the skeptic's *victory* "bloodless" or even "fictitious."¹⁵ He flatly concedes to any critic of skeptical or fideist bent that it is impossible to provide a rational justification of one's basic philosophical standards, principles, procedures—among which Ayer seems to regard "inductive inference" as one of the most important. It is, he says, impossible to give a proof "that what we regard as rational procedure really is so; that our conception of what constitutes good evidence is right."¹⁶

Whatever its faults, Ayer's approach does possess the initial advantage of honesty about the unjustifiability of ultimate standards and principles. By candidly discarding the claim that all legitimate positions must be rationally justifiable, he seems to avoid the problem of integrity, of claiming the ability to do more than he logically can. This is the main reason for calling his position a *critical* rationalism. If one's theory of rationality does not *require* one's standards of rationality to be rationally justifiable, then the skeptic's argument that it is *impossible* to do so loses its force even when it is admitted as correct.

It is not, of course, sufficient for Ayer simply to abandon the demand

¹⁴ Although the critical rationalism discussed here resembles terminologically the view Popper discusses under the same name in *The Open Society*, Chap. XXIV, and elsewhere, the problem he was discussing there is somewhat different from my own; his argument was concerned less to defeat the rational excuse for irrational commitment than to deal with more radical and—to my mind—less intellectually serious forms of irrationalism which were prevalent during the pre-war and wartime years when *The Open Society* was being conceived and written. I believe Popper would largely agree with the view I call "comprehensively critical rationalism," a view that I could hardly have stated without the help of his views.

¹⁵ A. J. Ayer, *The Problem of Knowledge*, pp. 75 and 80.

¹⁶ *Ibid.*, p. 74.

that his standards of rationality be justified. He must go on to show why his approach, *as a theory of rationality*, can dispense with the justifiability requirement in respect to standards. If Ayer could do this he would indeed thereby trivialize his admission that the skeptic succeeds in showing the impossibility of any such justification. The skeptic's argument about impossibility would "draw blood" from the rationalist only if the latter needed such justification.

As might be expected, Ayer's discussion conveys the impression that he is fully alert to the scope of his task. He writes of the importance of showing, "in a way that satisfactorily disposes of the skeptic's disproof," that the procedures "which sustain our claim to knowledge . . . do not *require* a proof of their legitimacy."¹⁷

But Ayer's response to his task, as contrasted with his appraisal of it, is curious and disappointing. Why do our standards of rationality not need rational justification? Simply, according to Ayer, because any such standard "could be irrational only if there were a standard of rationality which it failed to meet; whereas *in fact* it goes to set the standard: arguments are judged to be rational or irrational by reference to it."¹⁸ "When it is understood," Ayer explains, "that there logically could be no court of superior jurisdiction, it hardly seems troubling that inductive reasoning should be left, as it were, to act as judge in its own cause. The skeptic's merit is that he forces us to see that this must be so."¹⁹ Since there can be no proof that what we take to be good evidence really is so," then "it is not sensible to demand one."²⁰

There are several reasons why Ayer's position is unsatisfactory. Here I shall consider only the question of whether his viewpoint, even if assumed to be internally coherent, is successful *as a theory of rationality*. Unfortunately, the answer to this question is negative. The nub of the skeptical and fideistic objection was not, after all, simply the argument that comprehensive justification is impossible. It was, rather, that *since* comprehensive justification is impossible, the choice between competing ultimate positions is arbitrary. It is this argument, in its strongest form, that a successful theory of rationality would have to meet.

Viewed in this light, Ayer's argument falls to pieces. For (1) it begs the question; and (2) it can be shown to be itself a variety of fideism and hence no answer to it.

First, how does it beg the question? His argument is that our standards of rationality enjoy a special immunity from the justification demand since it would be impossible to judge these standards to be irrational.

¹⁷ *Ibid.* (my italics).

¹⁸ *Ibid.*, p. 75.

¹⁹ *Ibid.*

²⁰ *Ibid.*, p. 81.

These standards, Ayer argues, *set the standards on which any such judgment of their own irrationality would have to be based.*

Ingenious as it might seem, Ayer's argument could not be relevant to the present discussion, let alone valid, *unless* some particular standards and procedures of rationality, such as Ayer's own, which include "scientific induction," are *assumed* to be the correct ones. If some particular standards of rationality *are* the correct ones, then there can exist no other rational standards which are also correct but which can nevertheless invalidate the former as irrational. Yet this "if" marks a very big assumption; for *this is precisely what is at issue in the problem of rationality.* Whether regarded historically or theoretically, the most important criticisms of putative standards of rationality have questioned *whether* they were correct. Alternative conceptions of scientific argument and method, such as Popper's, which denies the *existence* of inductive procedure, let alone its rational legitimacy, do claim that there are standards of rationality which positions like Ayer's fail to meet.

Not surprisingly, a position that begs the question in this way turns out to be fideistic itself. Indeed, the "ultimate irrational commitment" of the fideist might alternatively be described as a self-conscious deliberate begging of the question. The main doubt about whether Ayer's position is fideistic would be whether in this case the question-begging is conscious and deliberate.

Ayer's position is fideistic in other respects. Apart from suggesting that any critic's demands would not be sensible, Ayer says rather little about how an unjustifiable rationalist position might be defended against a critic—whether a fellow philosopher-critic like Popper, or the most hyperbolic irrationalist—who simply does not "understand" why logically there can be no court of superior jurisdiction to arbitrate among standards like the principle of induction, whose rationality he does not *in fact* accept.

On this point Morton White has been more explicit. Although it might be asked whether Ayer fully shares White's position here, White has made it clear, in a review of *The Problem of Knowledge*, that he shares Ayer's. In the course of his review White warmly endorses Ayer's book and discusses some of its similarities to various philosophical movements in America, particularly pragmatism and its successors. When we want to defend something like the "general practice of basing our knowledge claims on experience," White suggests, "we can do no more than appeal to the *accepted code* for the transmissibility of the right to be sure . . . to the accepted way of speaking."²¹ To test the adequacy of his analyses, the philosopher can do no more than check them "against the moral convic-

²¹ Morton White, *Religion, Politics and the Higher Learning* (Cambridge: Harvard University Press, 1959), p. 47.

tions which he and others share.”²² Consequently the rationalist position, unable to be rationally based or justified, is finally based on irrational moral commitment. And the choice of this commitment is throughout dominated by *conservative* attitudes toward the best entrenched standards already accepted by one’s own philosophical community—indeed, just as conservative as those of the sixteenth- or seventeenth-century fideists who argued on similar grounds for adherence to the Church of Rome. How apt then—and how ironic—that in a different connection, in his book *Toward Reunion in Philosophy*, White should express his general preference of Erasmus to Luther. He writes:

Sometimes . . . the sanitation [i.e., removal of intellectual “refuse”] can be carried on with crusading enthusiasm, with a sense of deep antipathy to a philosophy which one opposes on every major issue, and which one would like to see extirpated. But at other times criticism can be conducted in the spirit of Erasmus rather than Luther, by one who admires a philosophical framework so much that he wishes to cleanse it of its shortcomings rather than demolish it. Much of the negative part of this book is intended in the second spirit, since I believe that the analytic, the empiricist, and the pragmatic movements . . . have been the most important and enlightening tendencies in twentieth-century philosophy.²³

How can one tell when to adopt the spirit of Erasmus and when that of Luther? If the final appeal is to the code and convictions of one’s colleagues, and if the view under consideration is one shared with them—indeed admired by them—then are the cards not stacked in favor of Erasmus? If White had lived during the Reformation and had practiced then the method he advocates in the twentieth century, would he not, like Erasmus, have remained within the fold of the Roman Catholic Church, fully critical of its shortcomings, but admiring its framework too much to want to see it extirpated?

One further, rather interesting point brings out more clearly the fideistic character of positions like Ayer’s and White’s. However radically their substantive opinions may differ, from a structural standpoint the positions of arch rationalists and antitheologians like Ayer and White are closely parallel not only, as might be expected, to fideistic positions like contemporary “Oxford theology,” but also that of the arch theologian and belligerently fideistic irrationalist Karl Barth, the leading Protestant Christian exponent of modern-day Calvinist “neo-orthodoxy.” In order to appreciate this striking parallel, it is necessary to note that as an alternative to the task of justifying rational standards Ayer sees the task of the philosopher as that of *describing* them—an important enough task if critical judgments are to be made by reference to the accepted procedures

²² *Ibid.*, p. 8.

²³ Morton White, *Toward Reunion in Philosophy* (Cambridge: Harvard University Press, 1956), pp. 289–90.

of rationality; to avoid errors we would need accurate descriptions of these procedures.²⁴ White agrees with Ayer that the business of the philosophically minded person is to analyse and to express in principles the patterns of the accepted ways of speaking or reasoning.

Interestingly enough, Barth argued that although the content of the "Word of God" might be expounded or described, it is neither possible nor necessary for the theologian to apologize for it, to defend it against skeptical criticism by trying to justify it. To do so is not only logically impossible anyway, but more strongly—since logical impossibility does not necessarily worry a theologian—it is unnecessary and irreverent for anyone genuinely committed to the "Word of God." There can be no assessment of the Word of God or of the Christian's ultimate commitment to it *since* the Word of God is itself *assumed* to be the standard or criterion which any such assessment would have to use.²⁵

In a similar vein, the theologian Paul Tillich blended fideism with a little "theological positivism" to write that "The assertion that something has sacred character is meaningful only for the asserting faith . . . There is no criterion by which faith can be judged from outside the correlation of faith."²⁶ Ayer did not put the point much differently: "Inductive reasoning should be left, as it were, to act as judge in its own cause."²⁷ In Ayer's case as in Barth's, when we confront that to which commitment has been made—just as when we read *Wind in the Willows*—"it is we who are being judged."²⁸

Indeed, the shift made by Ayer and others from comprehensive rationalism to critical rationalism can be described quite accurately in the traditional theological terminology. Both Ayer and the Christian fideists abandon *apologetic theology*, the kind of theology which tries to give rational justification for commitment, and replace it with *kerygmatic theology*, the theology devoted exclusively to the exposition and description of the fundamental message. Unable to justify his basic position, the logical empiricist, just like the neo-orthodox theologian, begins to describe the position, to preach it without regard to the critical situation within

²⁴ At this point Ayer's position links closely with many themes of the so-called Oxford philosophy of descriptive analysis, and also with the positions of many other leading American philosophers, such as W. V. Quine and Nelson Goodman. Goodman's unsuccessful but instructive attempts to describe the principles of inductive reasoning are chronicled in his *Fact, Fiction and Forecast*. For a discussion and criticism of Quine's holistic view, which radically differs from my own despite misleading similarities, see my book *The Retreat to Commitment*, where I discuss and reject his theory about the revisability of logic, and propose a "revisability criterion."

²⁵ See Barth's *Church Dogmatics*, particularly Vol. I, Part I, and my discussion of it in *The Retreat to Commitment*, Chap. III.

²⁶ Paul Tillich, *Dynamics of Faith* (New York: Harper & Brothers, 1957), pp. 58 ff.

²⁷ *The Problem of Knowledge*, p. 75.

²⁸ See the introduction by A. A. Milne to Kenneth Grahame's well-known book.

which it is being considered. As these basic similarities in their positions indicate, Ayer and like-minded rationalist philosophers beg the main question as flagrantly as do Barth and those who may share his commitments. While Barth begs the question of the existence and righteousness of God and his Word, Ayer begs the question of the existence and legitimacy of induction. And in common with various fideistic stances, Ayer's position bars in advance criticism of his fundamental standards by laying down what might be called a "persuasive definition" of "rational" in terms of which his own standards are automatically rational.²⁹

Presented with such a defensive definition, a critic can perhaps refuse to be persuaded by the prestigious title and ask whether, if what is being suggested is "being rational," it is right to be rational. He might also ask whether someone with a genuine concern to criticize and test his standards should not—instead of comparing them with the "accepted code" and the convictions of the group of which he is a member—seriously explore the views of those who are convinced the code is wrong but who may, for all that, be able to defend their ideas, and perhaps even be right. But more of this latter point in the following sections.

In sum, Ayer's position, his critical rationalism—like thousands of similar stands taken by contemporary philosophers—may indeed be one that can be held with considerable intellectual integrity, or with at least as much as that of any other fideistic position. But it is not a satisfactory theory of rationality. Indeed, it can only be held with integrity because it *fails* to solve the problem of rationality. To be satisfactory, a theory of rationality which, like Ayer's, began with the admission that the standards of rationality were unjustifiable, would have to go on to show—without begging the question—that the arbitrary irrationalism of skepticism and fideism could be escaped without comprehensive justification. This task Ayer's theory fails even to attempt.

VIII

Critical rationalism, like comprehensive rationalism, was doomed to failure by its structure; no future theory of rationality will succeed until this structure is, to use White's word, "extirpated." By the word "structure," I have in mind nothing more mysterious than certain features of these theories of rationality which predetermine the kinds of questions asked in philosophical discussion and limit the range of answers deemed appropriate. My attempt to bring this structure to light in its historical

²⁹ Herbert Feigl had already made the last point, though not in reference to Ayer, in a fascinating paper published some four years before Ayer's book. See Feigl's "Validation and Vindication: An Analysis of the Nature and the Limits of Ethical Arguments," in *Readings in Ethical Theory*, p. 676.

context, to break it, and to put forward an alternative begins firstly from an important historical observation or discovery about our traditional philosophical questions: namely, that *our philosophical tradition is authoritarian in structure, even in its most liberal forms*. My second step will be to show how two unfamiliar and unnoticed philosophical dogmas, assumed throughout our philosophical tradition without question, are responsible for this structure. These two dogmas are: (1) The assumption that criticism is necessarily fused with justification; and (2) The assumption that the quality and degree of rationality pass through the relationship of logical deducibility from justifying premises to justified conclusions.

The authoritarian structure of our philosophical questioning has been concealed from the attention of philosophers, and might still be doubted by some of them, because of the standard oversimplified way in which the rise of modern philosophy is described as part of a rebellion against authority. So it was—but a rebellion against *particular* traditional authorities, not against authority as such. Far from repudiating the very appeal to authority, modern philosophy has entertained seriously only one alternative to the practice of basing opinions on irrational and traditional authorities; namely, that of basing them on *rational authorities*.

The traditional questions of philosophy are authoritarian in structure in the sense that *they all beg authoritarian answers*. Questions like “How do you know?” “How do you justify your beliefs?” or “With what do you guarantee your opinions?” demand authoritarian answers, whether those authorities in particular cases be the Bible, the leader, the social class, the nation, the fortune teller, the Word of God, the intellectual intuition, or sense experience. A demand for a justification or a guarantee cannot be answered except by providing something authoritative in the sense that it is unquestionable, does not itself need justification, and hence can guarantee the correctness of a conclusion.

This historical observation I owe to an address Popper read to the British Academy in 1960: “On the Sources of Knowledge and of Ignorance.”³⁰ His simple but brilliant observation has, I believe, an almost revelatory character when the history and problems of philosophy are reconsidered in its light. I shall try to build on this observation, first by putting it in some philosophical context, then by explaining it, and finally by suggesting the principal outlines of my new theory of rationality—comprehensively critical rationalism—which can be erected within the new, roomier structure Popper’s observation makes possible.

But first we might ask whether Popper himself had an alternative

³⁰ Printed in *The Proceedings of the British Academy*, 1960, and published separately in 1961 by Oxford University Press (Henriette Hertz Trust monographs). The address is reprinted as the Introduction to *Conjectures and Refutations*. A preliminary statement of the view appears in Popper, “On the Sources of Our Knowledge,” *Indian Journal of Philosophy*, August 1959.

approach to suggest. What new questions would be asked in philosophy if the old ones were abandoned because of their authoritarian character? Indeed, is it really possible to forsake the old questions without abandoning philosophy? Is a nonauthoritarian theory of rationality possible?

I think the importance of Popper's observation should have become apparent already, before it has even been applied, in that the questions just posed, and most interestingly the last one, could not even *arise* until the observation about the authoritarian character of traditional philosophy had been made. Yet, as we shall see below, these new questions lead directly to the solution of the problem of rationality.

In explaining the implications of his observation, Popper not only rejected, as had Ayer, the demand for rational proofs of our rational standards; he went further, to suggest that the demand that *everything else except* the standards be proved by appealing to infallible intellectual authorities (which do not exist anyway) should also be abandoned. Instead of following Ayer in replacing philosophical justification by philosophical description, Popper urged the *criticism* of standards as the main task of the philosopher. Philosophers, he argued, should not demand and search for infallible intellectual authorities, but should instead try to build a philosophical program for counteracting intellectual error. Rather than search for an infallible intellectual authority to guarantee the truth of his views, the philosopher should seek a way of eliminating error from them. Within such a program, the traditional "How do you know?" question would not legitimately arise. And if it arose in fact, the philosopher would have to reply: "*I do not know*; I have no guarantees." To be more precise, such a philosopher might elaborate: "Some of the theories I hold may in fact be true; but since there are no criteria of truth, I can never know for sure whether what I believe to be true is in fact so." For such a philosopher a different question would become important: "How can our intellectual life and institutions be arranged so as to expose our beliefs, conjectures, policies, sources of ideas, traditional practices, and the like—whether justifiable or not—to maximum criticism, in order to counteract and eliminate as much intellectual error as possible?"

Coming from Popper, of course, a program of criticism is no surprise. In this context it is a very general extension of the familiar "criticize and try to falsify" approach he uses in the philosophy of science. Specific elements of his solution, including a discussion of the way philosophical questions acquire an authoritarian structure, are already to be found in his discussion of political leadership in *The Open Society*, Chapter VII. But the explicit link he establishes in the academy address between the more familiar parts of his theory of science on one hand, and the general structuring of philosophical inquiry on the other, is remarkably fresh and revealing, not only in regard to the solution of philosophical problems, but also in assessing Popper's own contributions, in setting his position in its

proper philosophical context, and in helping to explain why it is so often misunderstood.

I believe that the general shift from the demand for authoritative justification to the demand for criticism is a genuine innovation in philosophy—by no means limited to the “falsification versus verification” issue—whose importance can hardly be overemphasized. All the same, it might be objected that the “innovation” I have been heralding so confidently is imaginary; that the emphasis on the critical evaluation of competing views, far from being novel, is just another rather off-key refrain of the “hymn to criticism” which has been in the philosophical psalter since the pre-Socratics. Talk of the critical attitude tends to become boring when almost everybody fervently approves of it—or says he does. Indeed, I have several times heard philosophers deplore Popper’s “tiresome repetitiveness” about the critical attitude. “One hears it in every lecture. As if such things need to be said—particularly to philosophers!”

Although such reactions are often understandable, they are to be deplored. Such closing of the ears at the mention of the word “criticism” suggests that neither the words nor the music of Popper’s particular song have been heard. And this is a pity, for the notion of criticism, far from being trite, is one of the most unexplored, puzzling, and perhaps exciting and rewarding areas of philosophy. To show why such reactions are mistaken, and to explain why this particular kind of tone deafness is so prevalent, I wish to try to bring out as clearly as possible the crucial difference between the *new* idea of criticism I believe is being advocated here and the old familiar themes of traditional critical philosophies.

IX

This can be done fairly directly by asking for an explanation of Popper’s observation. *Why has an authoritarian structure been retained—and even gone unnoticed—in modern philosophies that have been intentionally and often explicitly antiauthoritarian and critical in spirit?*

The following assumption provides the most important element of the answer to this question: *The task of solving the problem of critical rational arbitration among competing positions has been frustrated by the fact that in almost all traditional and modern philosophies—those that have called themselves critical as well as those that have not—the idea of criticism has been fused with the idea of justification.*³¹ Since demands for justification are, of course, satisfied by the appeal to authority, the dilemma of

³¹ A number of historical positions doubtless approached toward a separation of justification and criticism. That Kant may have been doing so is suggested by his ambivalent use and description of the categorical imperative. However, I know of no philosophy before Popper’s that succeeded in separating the two.

ultimate commitment arises *in respect to criticism* in such philosophies. As a group, the philosophies in which this fusion of justification and criticism occurs may be called justificational philosophies of criticism.

The main originality of Popper's position lies in the fact that it is the first *nonjustificational philosophy of criticism* in the history of philosophy. This aspect of his thought may help explain why some misunderstandings of his position have been so persistent. Many of his critics, not surprisingly failing to see this point, have tried to deal with his thought within a philosophical structure and system of classification foreign to it, and have consequently led themselves to conclude that Popper's theory is subject to the same difficulties as its predecessors. I suspect that the fusion of justification and criticism in Ayer's thought may explain why he turned to description when justification broke down; for criticism would only appear as an *alternative* to justification after the two notions had been separated.

The purpose of the view advocated here is to help make future hymns to the critical attitude valuable by making *explicit* the nonjustificational character of Popper's critical thinking—an aspect which it may be especially useful to bring to attention since Popper has throughout his writings *practiced* nonjustificational criticism without explicitly discussing it in general terms.

But what does it mean to talk of the fusion of justification and criticism? Until this is explained, the importance of a separation of the two notions can hardly be assessed. They have been fused in a number of different ways. One of these, historically probably the most important, dominates most kinds of comprehensive rationalism. In this view, the way to criticize a view is to see whether it can be logically derived from, i.e., "justified by," the rational criterion or authority. In Hume's empiricist view, for instance, the strongest way to criticize a particular theory was to show that it could not be justified or established properly—in his case by an appeal to sense experience. On examining Hume's philosophical writings, one finds him making fairly consistent use of this basic strategy of criticism. He takes one idea after another—the idea of God, of the soul, of memory, of other minds—and asks whether it can be justified by being derived from sense experience, which he regards as man's only source of knowledge, or rational authority. If it can be justified as required, he accepts it; if not, he either rejects it or implies that from a rational point of view it should be rejected.

The analytic and synthetic procedures of Descartes's method "for conducting the reason well and for searching for truth in the sciences," however different in other respects, are closely parallel to Hume's in this. Ideas that cannot be reduced to clear and distinct ideas, and thus rationally justified, Descartes thinks should be rejected, just as everything that is to be accepted should be so justified. For both men, the rational way to criticize an idea is to see whether or not it can be rationally justified.

Another popular strategy of criticism, although weaker than the first in its demands, also fuses justification and criticism. It is weaker in that it employs a kind of "elastic clause" similar to that in the United States Constitution. What matters is not whether a belief can be derived from the rational authority but whether it conflicts with it. In other words, it is not irrational to hold a belief that cannot be derived from, i.e., justified by, the rational authority unless its denial can be derived from the rational authority. The second strategy can be varied in many quite subtle ways. Indeed, a typology of theories of authority, developed in terms of the different possible critical moves consistent with the general strategy, might illuminate some of the particular twists taken in historical controversies.³² Yet all varieties fuse justification and criticism in one way or another; to criticize a position, one must either show that it cannot be derived from, or else that it conflicts with, the rational authority, which is itself not open to criticism.

X

Since such an account of justificational strategies of criticism leaves undecided the character, or even the possibility, of a nonjustificational type of criticism, we need at least a rough picture of the latter type in operation, as well as some further clarification of the difference between justificational and nonjustificational theories. I think we can get both from trying to answer yet another question: *Why have justification and criticism repeatedly been fused in the ways described?*

In effect, this is a request for an explanation of the assumption just discussed, and may be obtained by bringing to light a further hidden philosophical dogma: *Most philosophical views have tacitly taken for granted that rational character and degree of rationality are properties that pass from premises to conclusion in the same manner as the property of truth through the relationship of logical deducibility.*³³ That is, most

³² For example, it might illuminate a number of controversies in the history of jurisprudence, such as the question of the province of, and the demarcation between, positive law and morality. See H. L. A. Hart, "Positivism and the Separation of Law and Morals," *Harvard Law Review*, February 1958, p. 599: "What both Bentham and Austin were anxious to assert were the following two simple things: first, in the absence of an expressed constitutional or legal provision, it could not follow from the mere fact that a rule violated standards of morality that it was not a rule of law; and conversely, it could not follow from the mere fact that a rule was morally desirable that it was a rule of law."

³³ Several philosophers of science, including Carnap, Hempel, Goodman, Popper, and Watkins, have in recent years—without discussing its general historical background—considered the place of a similar assumption in theories of "confirmation" in the natural sciences, referring to this assumption by names like "consequence con-

theories about the rational evaluation and criticism of competing views include (1) some more or less well-defined notion about the character of whatever standard of rationality or intellectual respectability is to be used in making evaluations and criticisms, and (2) the assumption that this intellectual respectability, whatever its nature, must be fully transmissible—like truth and unlike falsity (which is retransmitted)—from premises to conclusion through the deducibility relationship. According to the assumption, the logical derivatives of a theory inherit its quality and degree of intellectual respectability.

The historical origins of this assumption, as well as its unusual tenacity, are probably to be explained by the fact that the earliest attempted criteria of respectability were regarded as *criteria of truth*. In such views, the demarcation between the respectable and the disreputable coincides with the demarcation between the true and the false. These early attempts met many difficulties. Criteria of truth proved to be either unattainable or practically inapplicable to the issues for which they were needed; and the search for criteria of truth was gradually displaced—although never completely abandoned—by a search for some weaker but more attainable measures. The property most often used for this task, particularly by scientists and mathematicians, was that of probability. Prior to the development of modern probability theory several different senses of “probability” came to be used for this purpose.³⁴ With the development of the probability calculus, however, probability *in the sense of the probability calculus* has increasingly—but not universally!—been applied in attempts to provide the measure of rational respectability which would be the best substitute for a criterion of truth.

Truth and *probability* (to be used henceforth in the sense of the probability calculus) happen, however, to be two of the very few characteristics which are indeed transmissible from premises to conclusion through

dition,” “entailment condition,” and “content condition.” Although, as will be apparent, I agree with Popper’s position, I do not intend my remarks to apply only to scientific subject matters; the problem of rational evaluation, and the conditions under which it has been discussed, extend through the various subject matters of philosophy. For some earlier remarks about this assumption see my article, “A Note on Barker’s Discussion of Popper’s Theory of Corroboration,” *Philosophical Studies*, January-February 1961, p. 8. It is more accurate to read there “deducibility relationship” for “implication sign” since we are speaking metalinguistically. The idea of “condition sign” (Quine) or “implication sign” (Carnap) would coincide with this only in demonstrable relationships.

³⁴ There is, for example, the very interesting sense of probability Sextus Empiricus attributed, during the Hellenistic period, to Carneades. Sextus reports that Carneades was compelled to adopt a criterion for the conduct of life which made use of both “probable presentations” and those which are “at once probable and irreversible and tested.” Although it is hard to reconstruct from the texts exactly how this conception of probability operated, it could hardly be a type of probability in the sense of the probability calculus. See Sextus Empiricus, *Against the Logicians*, Book I, pp. 166 ff.

the deducibility relationship.³⁵ I suggest, as an historical conjecture, that these two concepts exerted such a determining influence over the early developments of the theory of rationality that it has been unquestioningly assumed that other putative measures and standards of respectability, however they might differ from truth and probability, would nevertheless automatically share their logical transmissibility. Indeed, the demand for justification made undesirable any measure unable to justify its derivatives by lending them its respectability.

The whole self-reinforcing structure of traditional philosophy so endorses and protects this assumption that even today, when criteria of truth are as unavailable as ever, and when probability measures have been found to be quite incapable of arbitrating rationally among competing scientific hypotheses, not to mention less precise ideas, the property of logical transmissibility is still expected of other evaluatory measures without regard to their real logical capabilities.

These remarks may help explain the persistence of some empiricists in attempting to retain probability as a measure despite devastating criticism. The transmissibility capacity of truth and probability led, historically, to the general assumption that intellectual respectability is transmissible from premises to conclusion. But historical origins are easily forgotten, so that the situation now seems ironically reversed; probability measures are retained *because* they are logically transmissible. The transmissibility requirement itself is taken for granted.³⁶

It is a matter of interest, if not surprise, that many theories that have relied on probability as a rationality measure have also included "empirical character" as a further requirement for rationality. In such a view legitimate statements would have to be truth-functionally reducible to something like individual "basic statements" reporting sense experience; and the logical derivatives of a legitimate statement would inherit not only its degree of probability but also its empirical character. Empirical character, however, turns out *not* to be transmissible through the deducibility relationship. From every basic empirical statement both nonempirical metaphysical statements and all tautologies follow logically. Adding to the

³⁵ See Popper, "The Demarcation Between Science and Metaphysics," *op. cit.*, footnote 63. Another such property would be "the right to be sure," as discussed and used by both Ayer and White. In saying that probability is transmissible, I mean of course that a consequence will be *at least as probable* as the premise from which it is derived. It may of course be *more* probable.

³⁶ This account does not, of course, fully explain such apparent anomalies as Carnap's simultaneous *rejection*, in *Logical Foundations of Probability*, p. 474, of the applicability of the transmissibility assumption or "consequence condition" to the theory of confirmation, and *acceptance* of the view that degree of confirmation coincides with degree of probability. Since probability is itself transmissible from premises to conclusion, Carnap's position would appear to be inconsistent.

difficulty, universal scientific hypotheses cannot be reduced to truth functions of a finite class of basic empirical observation statements—which denies empirical character to scientific hypotheses themselves. Such unwanted results hound empiricists with the well-known “paradoxes” of induction and confirmation in any case; but when transmissible probability is mixed with the nontransmissible property of “empirical character,” the results may be bizarre. Nonempirical consequences of empirical statements inherit the probability of the original empirical statement, and consequently become respectable from the point of view of the probability standard, but rather disreputable from the empirical standpoint. When “meaning criteria” are added as still further criteria of respectability, still more anomalies appear.³⁷

If it were really fundamentally similar to other critical philosophies, Popper’s theory of critical evaluation might be expected to share some of these perennial characteristics and difficulties. Actually, when we turn to the theory of testability he devised for evaluating scientific theories, we find an entirely different approach. Although the theory of testability is a genuine approach to the familiar problem of assessing and criticizing competing theories rationally, it does not contain the deducibility assumption and is, in fact, incompatible with it; and the criticism involved is entirely nonjustificational.

His theory of testability is an attempt to assess as one (but not the only) measure of rationality not the degree to which a theory is probabilified or confirmed or justified by evidence, but the degree to which it is testable or falsifiable and has been subjected to deliberate criticisms, tests, or attempts to falsify it. The measures of degree of testability or degree of corroboration, unlike truth and probability, are not logically transmissible from premises to conclusion. Quite the contrary, these properties share with *falsity* the property of *retransmission* from conclusion to premises. Consequently, the difference between Popper’s theory and the others referred to can be stated—and even defined—in the following terms. Whereas evaluational properties like probability and degree of confirmation are transmitted in the same direction as truth, degree of testability is retransmitted in the *opposite* direction like falsity. Thus the difference between the testability theory and various probabilistic theories of confirmation can be defined by reference to the irreducible difference in the ways the two concepts are logically transmitted.

If Popper’s theory of testability did share with rival theories the assumption that the measure of intellectual respectability flowed from premises to conclusions which they justified, then any logical consequence of an hypothesis would have to be as highly testable as the original hypo-

³⁷ Popper has of course discussed these many times. See for examples “The Demarcation Between Science and Metaphysics,” *op. cit.*

thesis: testability would be transmissible from premises to conclusion. But in fact the opposite happens: since a hypothesis is testable or falsifiable by the falsification of any of its consequents, the hypothesis must possess at least as high a degree of testability as any of its consequents. But it *may* possess a higher degree of testability—if, for example, it possesses other consequents independent of the first. And if the hypothesis can possess a higher degree of testability than its consequents, then the consequent does not inherit this particular evaluational property through the deducibility relationship. A high level, highly testable scientific theory does not, then, bequeath its degree of falsifiability or testability to those lower level theories it entails, its necessary conditions, which traditional accounts would represent it as justifying.

This point might be illustrated by examining the testability relations of these three hypotheses:

1. All who dwell in London are English.
2. All who dwell in Hampstead are English.
3. All who dwell in Bloomsbury are English.

Assuming, correctly, that Bloomsbury and Hampstead are both in London and that both the second and the third statements follow from the first, let us suppose that the second statement is falsified. By *modus tollens*, the rule of retransmission of falsity, the first statement is falsified too.

But suppose that the second statement has not been falsified, and that another of the first statement's consequents, the third statement, has never been tested. Clearly, the first statement will be falsified by the third just in case the third is subsequently tested and found false. But the second statement will not be falsified thereby, since it is logically unrelated to the third. Thus the first statement is more testable or falsifiable than the second, since the first is falsifiable by something that does not falsify the second. And the second cannot be more falsifiable than the first, since anything that falsifies the second falsifies the first.

This example, and similar ones, bring into relief not only the absence of any assumption that a measure of intellectual responsibility or rationality (in this case "degree of testability") is logically transmissible from premises to conclusion, but also the nonjustificational character of Popper's theory of criticism.

To avoid confusion of this approach with the second strategy of *justificational* criticism discussed in the previous section, it should be noted that in the theory of corroboration the falsity of a view is not *established* in a refutation thereof. Rather, the view is provisionally rejected because it conflicts with some other better tested, less problematic view. But the view that occasions the refutation is itself open to criticism by the testing of its own consequences. And these in turn are criticizable; and so on forever. This *process* of testing is, of course, in principle infinite; but there is no

infinite *regress*, because the aim of justifying or establishing has been abandoned.³⁸

If all measures of degree of intellectual respectability resembled truth and probability in passing from premise to conclusion through the relationship of logical deducibility, all criticism would have to be justificational. The very fact that degree of corroborability and testability are not such qualities establishes not only that not all criticism need be justificational, but also that this type of criticism *cannot* be justificational. Hence a nonjustificational, nonauthoritarian theory of knowledge and rationality is certainly possible.³⁹

The features of traditional philosophy discussed in the last three sections—the authoritarian structuring of its questions, and the two dogmas that are largely responsible for this structuring—each deserve further detailed exploration and explanation, as does the important, more or less technical question of how to construct critical institutions in philosophical inquiry. I hope only to have suggested where some profitable investigations may lie. But even in their present form I believe the matters just discussed are sufficient to suggest immediately the character of a theory of rationalist identity that can escape the dilemma of ultimate commitment and thereby avoid both skepticism and fideism.

XI

The new concept of rationalist identity I have in mind is already implicit in the nonjustificational approach just sketched. I suggest that within

³⁸ See in this connection Popper, *Logic of Scientific Discovery*, p. 105. It might be objected that this passage from the *Logic of Scientific Discovery* shows that Popper did not at the time he wrote that book hold a nonjustificational theory of criticism. For he writes on page 105: "The basic statements at which we stop . . . have admittedly the character of *dogmas* . . . in so far as we may desist from justifying them by further arguments (or by further tests)." The word "justifying" here does not, however, support this objection. It is not a remnant in Popper's writing of justificational *philosophy* (which is inconsistent with the position he expounds in the book—see for example Appendix *9, *Logic of Scientific Discovery*); rather it is a remnant of the *terminology* of justificational philosophy. This could hardly have been otherwise at a time when the basic contrast between justification and criticism had not yet been made explicit. I tried to do this when writing my book; and during Popper's seminar at the London School of Economics in October 1961 I suggested the distinction to him in an attempt to explain why certain aspects of his thought had so often been misunderstood, and in part to excuse and defend some of his more persistent critics. By pointing up the incorrect but understandable assumptions on which the critics had based their interpretations of and objections to his thinking, I suggested that some of their arguments, although indeed incorrect, deserved more credit for their intelligence and seriousness than is sometimes accorded them.

³⁹ I am indebted to Professor Popper, and to Dr. Jerzy Giedymin, for correspondence and conversation which helped me to correct and improve this section—although I am not sure that they would agree with my final formulation.

this new framework the rationalist identity might be characterized as that of one who holds *all* his beliefs, including his standards and his basic philosophical position itself, open to criticism; one who protects nothing from criticism by justifying it irrationally. The position, which I call comprehensively critical rationalism, has several affinities with comprehensive rationalism: it shares its comprehensive aims, and it follows from, or is implied by it. That is, any comprehensive rationalist who succeeded *per impossible* in justifying *all* his opinions rationally would clearly not have to justify any of them irrationally.

Nevertheless the two concepts differ considerably. For instance, if they were equivalent positions, the traditional requirement would also be implied by the new one—which it is not. Significantly, it does not follow that a man who justifies none of his beliefs irrationally will be able to justify all of them rationally. If the aim of comprehensive justification has been abandoned he need not justify some, or even most, of them at all.

The new concept of rationalist identity also differs from critical rationalism, wherein the rationalist accepts the *rational* unjustifiability of his position but goes on to justify it in an *irrational* commitment to principles that are not themselves open to criticism.

If rationality lay in justification, such eventual irrational justification or commitment might seem inescapable. However, if rationality lies in criticism, and if we can subject everything to criticism and continuing test, including the rationalist way of life itself, then rationality is in this important sense unlimited. If all justification—rational as well as irrational—is truly abandoned, there is no need to justify irrationally a position that is rationally unjustifiable. The position may be held rationally regardless of justification, provided it can be, and is, held open to criticism and survives severe testing. The question of how well a position is justified is quite different from the question of how well a position is criticized.

The general separation of justification and criticism *can*, of course, be extended to the examination of the rationalist position itself. Whereas under traditional conceptions of rationalism the rationalist position was itself not rational and thus perpetuated crises of integrity, under the new conception the crisis disappears. The new rationalist identity satisfies its own requirements; without any contradiction or other difficulty the practice of critical argument can be criticized. Just as it is possible for a democracy, during the democratic process of voting, to commit suicide, so a comprehensively critical rationalist, who is not committed to the view that his position is the correct one, could be argued out of rationalism by himself or someone else. Fideistic irrationalists have, by contrast, argued that, even without actually abandoning allegiance to the object of commitment, merely subjecting the allegiance to criticism is to forsake the allegiance. For a comprehensively critical rationalist, continued subjection to criticism of his allegiance to rationality is explicitly part of his rationalism.

This spells defeat for the *tu quoque* argument that supports the skeptical and fideistic claim. The case for irrational commitment, as well as the rational excuse for it, rests on the argument that rationality is so limited logically that such commitment is inescapable. For the comprehensively critical rationalist there is no such limitation. If he accuses his opponent of protecting some belief from criticism through irrational commitment to it, he is not open to the charge that he is similarly committed. Consequently, the *tu quoque* argument cannot be used against comprehensively critical rationalism, and criticism of commitments no longer boomerangs.

XII

Since antiquity, irrationalists have embraced as fact—indeed as one of the few facts established in philosophy—the philosophical contention that rationality is logically limited; that at least insofar as his standards are concerned, every man must make some irrational commitment. They have often used this contention to excuse on rational grounds their own particular irrational commitments, however bizarre in character. Thereby they have been able in principle, though not in practice, to avoid loss of intellectual integrity. I have tried in the preceding pages to refute the philosophical theory about the limits of rationality by turning the tables, by showing how we can shift the emphasis in rational discussion from justification to nonjustificational criticism. I hope in this way to have contributed to a clarification of the issues of identity and integrity in the rationalist tradition. If my argument is sound, irrationalists lose the most formidable weapon in their intellectual armory, their rational excuse for irrational commitment. Those who continue as irrationalists will really be so, in the sense that they can no longer retain their various irrationalist identities while preserving the intellectual integrity the *tu quoque* has so long afforded them.

The Elimination of Variables by Regular Combinators

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